Lab 6: Code and Branch Coverage Analysis

Hospital Triage Priority (hospital_triage_priority.py)

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1. Objective

To measure line (code) coverage and branch coverage of the hospital triage module, identify untested paths, and write test cases that reach at least 90% for both.

2. Module Under Test

- process_triage_wards(wards, default_acuity_score): validates input, skips wards not accepting patients, flags invalid vitals, scores unconscious and abnormal-vitals patients, adds an ICU standby bonus, and returns WARD_ALERT, NO_URGENT_CASES or TRIAGE_STABLE.
- monitor_oxygen_manifolds(sensor_streams): returns NO_DATA, PRESSURE_FAILURE (two or more consecutive readings below 4.0 bar) or PRESSURE_NORMAL.

The module has 45 statements and 32 branches.

3. Initial Run (Starter Test)

The starter file test_hospital_triage.py contains one test (a single normal patient). Running it gave:

Measure	Result	Missed
Line coverage	47%	24 of 45 statements
Branch coverage	43%	8 branches partially covered

```
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage erase
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage run -m pytest test_hospital_triage.py
>>
==================== test session starts ====================
platform win32 -- Python 3.12.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47
collected 1 item

test_hospital_triage.py .                                  [100%]

==================== 1 passed in 0.06s ====================
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage report -m
Name                          Stmts   Miss  Cover   Missing
-----------------------------------------------------------
hospital_triage_priority.py      45     24    47%   3, 10, 21-22, 25-29, 31, 36, 39, 41, 46-58
test_hospital_triage.py           7      0   100%
-----------------------------------------------------------
TOTAL                            52     24    54%
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage erase
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage run --branch -m pytest test_hospital_triage.py
==================== test session starts ====================
platform win32 -- Python 3.12.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47
collected 1 item

test_hospital_triage.py .                                  [100%]

==================== 1 passed in 0.06s ====================
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47> python -m coverage report -m
Name                          Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------
hospital_triage_priority.py      45     24     32      8    43%   3, 10, 21-22, 25-29, 31, 36, 39, 41, 46-58
test_hospital_triage.py           7      0      0      0   100%
-------------------------------------------------------------------------
TOTAL                            52     24     32      8    48%
(.venv) PS C:\SE\SE_LAB_PES1UG24AM410_Sunidhi_Shekar_LAB\PES1UG24AM410_SELAB6_47>
```

Figure 1: Initial run of the starter test (line 47%, branch 43%)

Only the normal-patient path ran. Input validation, non-accepting wards, invalid vitals, unconscious patients, abnormal vitals, ICU standby, the WARD_ALERT and NO_URGENT_CASES statuses and the whole oxygen monitor function were untested.

4. Code Coverage Analysis

File: CodeCoverageTest_hospital_triage_priority.py (15 tests). Command used:

```
python -m coverage run -m pytest CodeCoverageTest_hospital_triage_priority.py
```

```
python -m coverage report -m
```

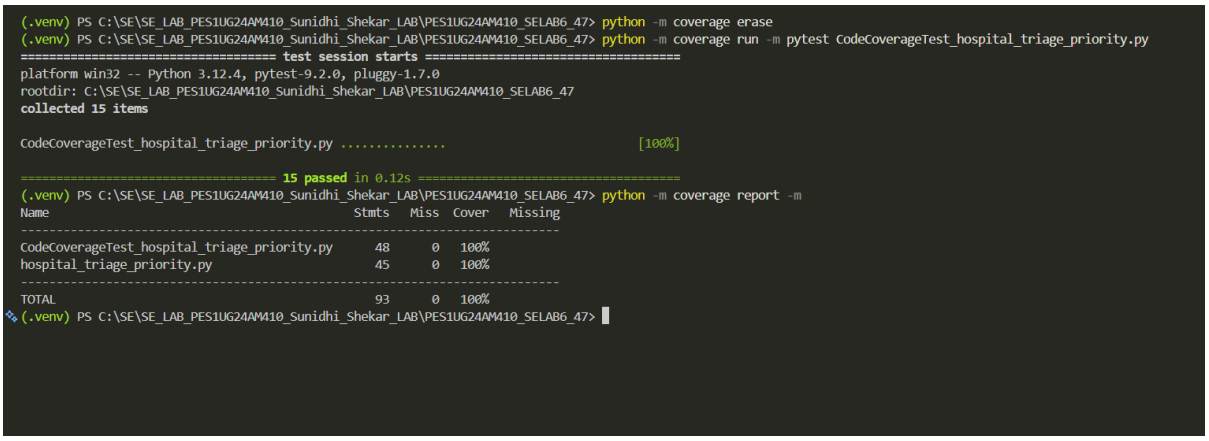


Figure 2: Line coverage, 15 tests passed, 100%

Result: 15 tests passed and hospital_triage_priority.py reached 100% line coverage (45 of 45 statements, 0 missed), up from 47%.

Tests were added for each previously missed group of lines:

Test	**Line(s) / path exercised**
invalid wards not a list; negative default score	ValueError raise
ward not accepting	continue for skipped ward
invalid vitals (spo2 = 0, hr = 0)	flagged entry and continue
unconscious, general ward	+100 acuity
unconscious, outpatient ward	score reset and break
spo2 = 85; hr = 130	+60 acuity
ICU standby	+25 acuity
five invalid entries	WARD_ALERT
no accepting patients	NO_URGENT_CASES
empty stream; two low readings; reset streak	NO_DATA, PRESSURE_FAILURE, PRESSURE_NORMAL

5. Branch Coverage Analysis

File: BranchCoverageTest_hospital_triage_priority.py (24 tests). Command used:

```
python -m coverage run --branch -m pytest
BranchCoverageTest_hospital_triage_priority.py
```

```
python -m coverage report -m
```

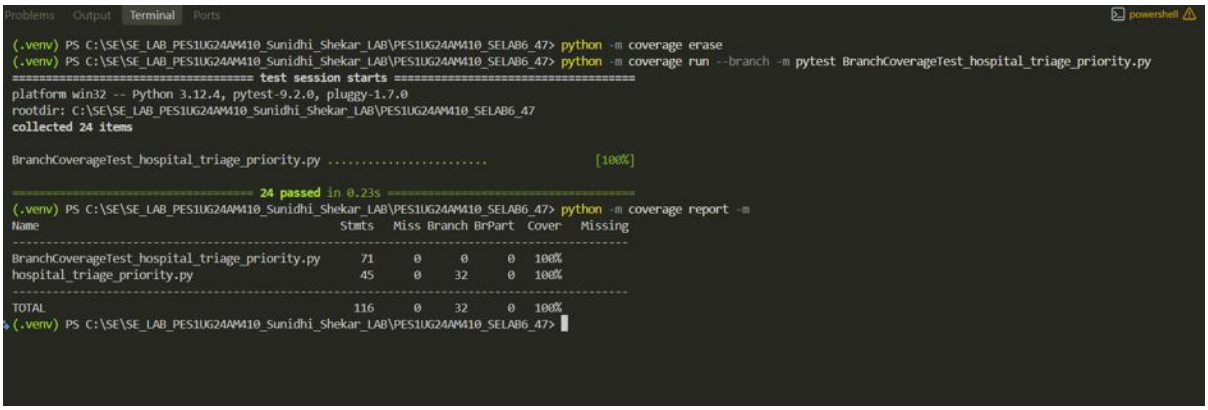


Figure 3: Branch coverage, 24 tests passed, 100%

Result: 24 tests passed and hospital_triage_priority.py reached 100% branch coverage (32 of 32 branches, 0 partial), up from 43%.

Branch coverage requires every decision to be taken both ways, so tests were added for the other side of each condition and for boundary values:

- default_acuity_score = 0 (the < 0 check is False)
- spo2 = 90 and hr = 120 (the exact boundaries are treated as normal)
- exactly 4 flagged entries (not an alert) versus 5 (WARD_ALERT)
- outpatient ward with a conscious patient
- ward with no patients and no icu_standby key
- a single low pressure reading (no failure), pressure exactly 4.0, and a failing second manifold

6. Line vs Branch Coverage

	Line coverage	**Branch coverage**
Question asked	Was each line executed at least once?	Did each if/loop decision go both True and False?
Initial	47%	43%
Final	100%	100%
Command	coverage run -m pytest	coverage run --branch -m pytest

7. Commits

Commit	**Message**	**Files**
9766c6a	Add code coverage test cases and screenshot	CodeCoverageTest_hospital_triage_priority.py, code_coverage.png
a68b023	Add branch coverage test cases and screenshot	BranchCoverageTest_hospital_triage_priority.py, branch_coverage.png

8. Conclusion

Coverage rose from 47% (line) and 43% (branch) to 100% for both, using 15 and 24 tests respectively. Coverage shows that every line and branch was executed, not that the logic is correct. For example, the break for an unconscious outpatient only exits the inner patient loop, so the ICU bonus and later wards still add to the score; coverage alone would not reveal whether that is intended.